# A Counterexample to Green's Problem 57

Working note

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**Abstract**

Problem 57 in Ben Green's *100 Open Problems* asks whether two notions of quadratically structured function coincide. We give an explicit negative answer on the smallest nontrivial odd cyclic group. For $G = \mathbb{Z}/3\mathbb{Z}$, a concrete modulus-one generalized convolution is separated from the restricted convex set by the real functional with coefficients $(20, -14, 5)$. Its unnormalized score is greater than 309.133, whereas every restricted function has score at most 308.73. The restricted bound is certified by three exact $6 \times 6$ positive-definite matrices over $\mathbb{Q}(e^{2\pi i/3})$; only three matrices are needed because of the symmetries of the 27 possible Fourier-index patterns.

## 1 The question and the answer

Let $G$ be a finite abelian group. Green defines $\Phi(G)$ to be the convex hull of the functions

$$\phi(g) = \mathbb{E}_{x_1+x_2+x_3=g} f_1(x_2, x_3) f_2(x_1, x_3) f_3(x_1, x_2), \qquad \|f_i\|_\infty \le 1. \tag{1}$$

The set $\Phi'(G)$ is defined in the same way, except that $f_3(x_1, x_2)$ is required to depend only on $x_1 + x_2$. Plainly $\Phi'(G) \subseteq \Phi(G)$. Problem 57 asks whether equality always holds [1].

**Theorem 1.** *For $G = \mathbb{Z}/3\mathbb{Z}$ one has*

$$\Phi'(G) \subsetneq \Phi(G).$$

*In particular, Problem 57 has a negative answer.*

All indices in the proof are taken modulo 3.

## 2 A separating functional

Put

$$h(0) = 20, \qquad h(1) = -14, \qquad h(2) = 5,$$

and consider the real linear functional

$$L(\phi) = \mathrm{Re} \sum_{g \in G} h(g)\phi(g).$$

For $G = \mathbb{Z}/3\mathbb{Z}$, the conditional average in (1) contains nine terms. Thus, for an elementary function in either convex set, $9L(\phi)$ is the unnormalized score

$$T = \mathrm{Re} \sum_{x_1, x_2, x_3 \in G} h(x_1 + x_2 + x_3) f_1(x_2, x_3) f_2(x_1, x_3) f_3(x_1, x_2). \tag{2}$$

We exhibit an unrestricted choice with $T > 309.133$ and prove that every restricted choice has $T \le 308.73$.

## 3 The unrestricted witness

Let $\zeta = e^{2\pi i/48}$. Define $f_r = \zeta^{E_r}$ entrywise, with $f_1$ indexed by $(x_2, x_3)$, $f_2$ by $(x_1, x_3)$, and $f_3$ by $(x_1, x_2)$, where

$$E_1 = \begin{pmatrix} 43 & 35 & 38 \\ 1 & 12 & 9 \\ 47 & 20 & 4 \end{pmatrix}, \qquad E_2 = \begin{pmatrix} 13 & 40 & 30 \\ 6 & 5 & 36 \\ 7 & 16 & 35 \end{pmatrix}, \qquad E_3 = \begin{pmatrix} 41 & 8 & 37 \\ 21 & 31 & 46 \\ 46 & 42 & 12 \end{pmatrix}.$$

Every entry has modulus one. Collecting the 27 terms in (2) gives

$$\begin{aligned} T = 30 + 147\cos\frac{\pi}{24} + 62\cos\frac{\pi}{12} + 40\cos\frac{\pi}{8} + 15\cos\frac{5\pi}{12} \\ - 10\cos\frac{11\pi}{24} + 14\cos\frac{\pi}{4} + 28\cos\frac{\pi}{6}. \end{aligned} \tag{3}$$

The elementary rational bounds

$$\begin{gathered} \cos(\pi/24) > .991, \quad \cos(\pi/12) > .965, \quad \cos(\pi/8) > .923, \quad \cos(5\pi/12) > .258, \\ \cos(11\pi/24) < .131, \quad \cos(\pi/4) > .707, \quad \cos(\pi/6) > .866 \end{gathered} \tag{4}$$

imply directly that

$$T > \frac{309133}{1000}. \tag{5}$$

For completeness, (4) follows by squaring the standard radical formulae. For example,

$$\cos^2\frac{\pi}{24} = \frac{4 + \sqrt{6} + \sqrt{2}}{8}, \qquad \cos^2\frac{11\pi}{24} = \frac{4 - \sqrt{6} - \sqrt{2}}{8},$$

together with $1.414 < \sqrt{2} < 1.415$ and $2.449 < \sqrt{6}$; the other five checks are even shorter. Numerical evaluation of (3) is $309.3102071492\ldots$.

## 4 The restricted upper bound

Suppose now that $f_3(x_1, x_2) = c(x_1 + x_2)$, and write $c_s = c(s)$. For fixed $x_3 = z$, introduce the $3 \times 3$ matrix

$$H_z(x, y) = h(x + y + z)c_{x+y}.$$

The vectors supplied by $f_1(\cdot, z)$ and $f_2(\cdot, z)$ have Euclidean norm at most $\sqrt{3}$. Their contribution to (2) therefore has modulus at most $3\|H_z\|_{\mathrm{op}}$. The matrix $H_z$ is reverse circulant. If $\omega = e^{2\pi i/3}$, Fourier diagonalization gives

$$\|H_z\|_{\mathrm{op}} = \max_{j \in \mathbb{Z}/3\mathbb{Z}} \left| \sum_{s \in G} h(s + z)c_s\omega^{js} \right|. \tag{6}$$

Indeed, multiplying $H_z$ by the permutation $y \mapsto -y$ turns it into an ordinary circulant matrix.

Choose a maximizing index $j_z$ in (6), write $\mathbf{j} = (j_0, j_1, j_2)$, and define

$$A_{\mathbf{j}}(z, s) = h(s + z)\omega^{j_z s}.$$

It remains to bound $\|A_{\mathbf{j}}c\|_1$. Choose phases $d_z$ so that

$$\|A_{\mathbf{j}}c\|_1 = \mathrm{Re}(d^* A_{\mathbf{j}}c), \qquad |d_z| \le 1,$$

and set

$$C_{\mathbf{j}} = \frac{1}{2}\begin{pmatrix} 0 & A_{\mathbf{j}}^* \\ A_{\mathbf{j}} & 0 \end{pmatrix}. \tag{7}$$

If $y_1, \ldots, y_6 > 0$ satisfy

$$D_{\mathbf{j}} - C_{\mathbf{j}} \succeq 0, \qquad D_{\mathbf{j}} = \operatorname{diag}(y_1, \ldots, y_6), \tag{8}$$

then, on applying (8) to the vector $(c, d)$,

$$\|A_{\mathbf{j}} c\|_1 \le \sum_{i=1}^{6} y_i. \tag{9}$$

**Lemma 1.** *To prove (9) for all $\mathbf{j} \in (\mathbb{Z}/3\mathbb{Z})^3$, it is enough to treat*

$$(0,0,0), \qquad (0,0,1), \qquad (0,1,2).$$

*Proof.* Adding a common constant $k$ to the three $j_z$ right-multiplies $A_{\mathbf{j}}$ by the unitary diagonal matrix $\operatorname{diag}(1, \omega^k, \omega^{2k})$. Negating all $j_z$ complex conjugates the problem. Finally, if $j'_z = j_{z+r}$, then

$$A_{\mathbf{j}'}(z, s) = \omega^{r j_{z+r}} A_{\mathbf{j}}(z + r, s - r),$$

so a cyclic shift changes $A_{\mathbf{j}}$ only by row and column permutations and unit-modulus row factors. These operations preserve $\sup_{|c_s| \le 1} \|A_{\mathbf{j}} c\|_1$ and transport a diagonal certificate without changing the sum of its entries.

If all three indices agree, the first representative results. If exactly two agree, the common value can be made zero, the exceptional value can be made one by negation, and its position can be moved by a cyclic shift. If all are distinct, an affine change $j \mapsto \pm j + k$ realizes every permutation of $(0, 1, 2)$. This gives the three stated representatives. $\square$

For the three representatives, the following exact choices satisfy (8). The coordinates are ordered as $(c_0, c_1, c_2, d_0, d_1, d_2)$.

| $\mathbf{j}$ | $100(y_1, \ldots, y_6)$ | $\sum_i y_i$ |
|---|---|---|
| $(0,0,0)$ | $(1478, 1478, 1478, 1478, 1478, 1478)$ | $2217/25$ |
| $(0,0,1)$ | $(1772, 1926, 1448, 1542, 1786, 1817)$ | $10291/100$ |
| $(0,1,2)$ | $(1713, 1713, 1713, 1713, 1713, 1713)$ | $5139/50$ |

Consequently (9) is at most $10291/100$ for every phase pattern. Combining this with (6) and the factor 3 from the two Euclidean norms gives, for every restricted elementary function,

$$T \le 3 \cdot \frac{10291}{100} = \frac{30873}{100}. \tag{10}$$

The same upper bound holds for every convex combination of restricted elementary functions.

## 5 Exact verification of the certificates

All entries of $D_{\mathbf{j}} - C_{\mathbf{j}}$ lie in $\mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. The matrices are Hermitian, so Sylvester's criterion applies. In the only case with a nonconstant diagonal certificate, $\mathbf{j} = (0, 0, 1)$, direct exact elimination gives the six leading principal minors

$$\frac{443}{25}, \quad \frac{426609}{1250}, \quad \frac{77216229}{15625},$$
$$\frac{105027914611}{3125000}, \quad \frac{8430332160187}{39062500}, \quad \frac{241353436796433}{7812500000}.$$

They are all positive. The same elimination for the two constant-diagonal cases gives six positive rational minors in each case. Thus all three matrices in (8) are positive definite. The accompanying checker prints all 18 minors and verifies their positivity using exact arithmetic; in particular, the restricted estimate is independent of floating-point optimization.

## 6 Conclusion

The witness from Section 3 belongs to $\Phi(\mathbb{Z}/3\mathbb{Z})$ and, by (5), has unnormalized score greater than 309133/1000. Equations (9)–(10) show that every element of $\Phi'(\mathbb{Z}/3\mathbb{Z})$ has score at most 30873/100. The certified separating margin is therefore at least

$$\frac{309133}{1000} - \frac{30873}{100} = \frac{403}{1000} > 0.$$

Hence the witness cannot lie in $\Phi'(\mathbb{Z}/3\mathbb{Z})$, proving the theorem.

## Reproducibility and status

The accompanying file `problem57_counterexample_check.py` collects the 27 witness terms, verifies the rational trigonometric bounds, and checks all 18 leading principal minors exactly in $\mathbb{Q}(\omega)$. It uses only the Python standard library. Numerical search was used to locate the witness and dual certificates, but no numerical approximation is used in the proof.

A targeted online search on 20 July 2026 located Green's statement of the problem but no prior resolution. This is a preliminary novelty check, not an exhaustive priority claim; corrections or earlier references would be very welcome.

## References


[1] Ben Green, *100 Open Problems*, Problem 57, 2026.
`https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf`